Chiral three-nucleon forces and pairing in nuclei

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Introduction

Pairing phenomena in nuclei

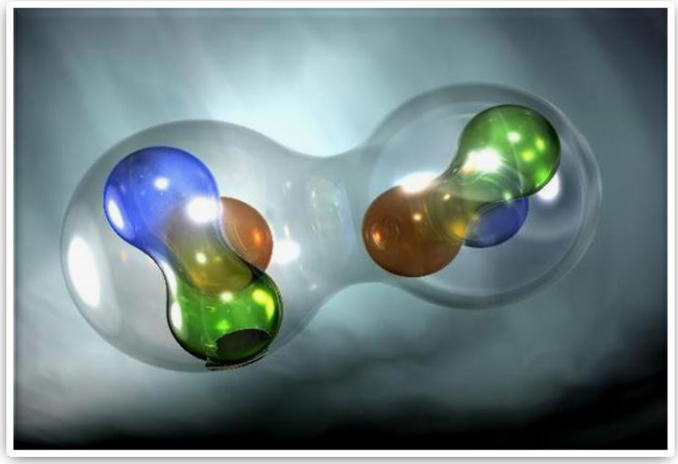


FIG.1. A deuterium nucleus, showing a kind of "pairing" between the two nucleons.

(Ref. https://physics.aps.org/articles/v14/155)

- In 1911 Kamerlingh Onnes discovered superconductivity in condensed matter systems, and its microscopic explanation came about through the highly successful pairing theory proposed in 1957 by Bardeen, Cooper, and Schrieffer (BCS) [1]
- Like-particle pairing is an essential aspect of nuclear structure, in particular regarding the description of exotic nuclei. [2]
- Pairing in infinite matter impacts the cooling of isolated neutron stars. [3]

Ref. : [1] Cooper et al., 1957 [2] J. Dobaczewski, W. Nazarewicz, Prog. Theor. Phys. Suppl. 146 (2003) 70 [3] S.K. Bogner, From low-momentum interactions to nuclear structure

Introduction

- The studies of pairing gaps from internucleon interactions for medium-mass and heavy nuclei, the method of especial choice is the nuclear energy density functional (EDF) approach
- EDF parameterizations provide a satisfactory description of low-energy properties of known nuclei
- Problem : EDF are empirical in character and lack predictive power as one enters experimentally unexplored regions.

- construction non-empirical EDFs

(deriving from microscopic nuclear forces.)

Purpose

- The study of pairing in nuclei including three-nucleon forces.
- The development of low-momentum interactions based on renormalization group (RG) methods

The RG plays a central role in EFT.

It determines the running of low-energy couplings

Introduction

What is the nuclear energy density functional (EDF) approach ?

Theoretical tool of choice for the microscopic and systematic description of medium- and heavy-mass nuclei.

This method has been empirically adapted from well-defined wave-function- and Hamiltonian-based approaches.

The EDF method aims at providing,

with one consistent frame,

(i) the detailed and complete description of specific nuclei of interest.

(ii) systematic trends over a large set of nuclei.

(iii) trustful extrapolations in the region of the nuclear chart where experimental data are and will remain unavailable.

Ref: [4] https://www.euroschoolonexoticbeams.be/files/nlp/LNP879_Chapter7.pdf

Historical perspective

- The Theory of Nuclear Forces: Seven Decades of Struggle

1935	**Yukawa: Meson Theory**
1950's	*The "Pion Theories".* One-pion exchange: good; multi-pion exchange: disaster.
1960's	Many pions ≡ multi-pion resonances: $\boldsymbol{\sigma}$, $\boldsymbol{\rho}$, $\boldsymbol{\omega}$, ... The One-Boson-Exchange Model: success.
1970's	Divers two-pion-exchange models: Partovi-Lomon, Stony Brook, Paris, Bonn.
1980's	Nuclear physicists discover **QCD**: Quark Models.
1990's and beyond	Nuclear physicists discover **EFT**; Weinberg, van Kolck, ... **Back to Meson (Pion) Theory!** *But, constrained by* $\boxed{\textit{Chiral Symmetry.}}$

- A major breakthrough occurred when the concept of an effective field theory (EFT) was introduced and applied to low-energy QCD.

- According to face the problem that the derivative coupling is not renormalizable in the conventional sense.

i) How to implement chiral symmetry consistently in the theory of pionic and nuclear interactions

ii) How to deal with the renormalization issue.

?

Weinberg's Folk Theorem is the stepping stone to what has become known as an Effective Field Theory (EFT).

- QCD at low energy is non-perturbative and, therefore, not solvable analytically in terms of the "fundamental" degrees of freedom (dof); quarks and gluons are ineffective dof at low energy.
- Below the chiral symmetry breaking scale $\Lambda_\chi \approx 1\ \mathrm{GeV}$ pions and nucleons are the appropriate dof.
- Moreover, pions are Goldstone bosons which typically interact weakly. Thus, there is hope that a perturbative approach might work.
- symmetry, such that our chiral EFT can be indentified with low-energy QCD, To ensure the connection with QCD, we must observe the same symmetries as low-energy QCD, particularly, spontaneously broken chiral D.

FIG. 2. Steven Weinberg at the American Physical Society Meeting in 1977.

the most general possible Lagrangian including all higher-derivative terms that are consistent with chiral symmetry → Results consistent with the underlying theory

Ref: [5] R. Machleidt, Nuclear Forces - Lecture 5NF from EFT I (Pohang 2016)

Bird's eye view of this work

I) Describing pairing within a perturbative expansion around the HFB state translates into solving a generalized gap equation.

input
- The normal self-energy (includes interactions between a single nucleon and the medium)
- The anomalous self-energy (computing from the pairing interaction kernel)

II) Systematic EDF calculations of semi-magic nuclei. (due to normal self-energy)

III) Systematic calculations of the odd-even mass staggering generated

(Using a microscopic pairing interaction at first order in chiral low-momentum interactions.)

IV) The studying of cutoff dependence of the pairing gaps.

(Given estimation of the theoretical uncertainties due to short-range higher-order NN and many-body interactions.)

Two-nucleon interactions (2NFs) from ChEFT

Pion-exchange (PE) contributions in ChEFT. :

Organize the contributions in terms

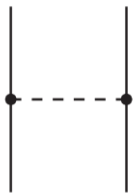
$(Q/\Lambda_\chi)^\nu$

⇩

"Power Counting"

Leading order

$(\mathrm{LO}), \boldsymbol{\nu} = \mathbf{0}$



Next-to-leading order

$(\mathrm{NLO}), \boldsymbol{\nu} = \mathbf{2}$

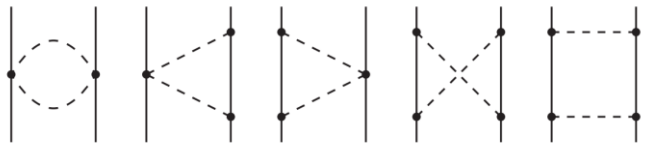


FIG. 4. One-pion exchange contributions. Diagram (a) represents the leading-order, while graphs (b)-(i) contribute to renormalization.

There are infinitely many pion-exchange contributions to the NN interaction, thus. We need to get organized.

At leading order $(\mathrm{LO}, \nu = 0)$

- On-shell and in the center-of-mass system (CMS), there are no relativistic corrections at any order.
- Off-shell corrections come into play when the relativistic 1PE is iterated.

⬇

At second order $(\mathrm{NLO}, \nu = 2)$

- The 1PE gets renormalized due to one-loop graphs and counter term insertions.

Ref. [6] R. Machleidta, D. R. Entem, Chiral effective field theory and nuclear forces

Two-nucleon interactions (2NFs) from ChEFT

Two-pion exchange (2PE) : The exchange of two or more pions always involves loop diagrams, which implies that we are faced with a non-trivial problem.

Next-to-next-to-leading order (NNLO), $\boldsymbol{\nu = 3}$

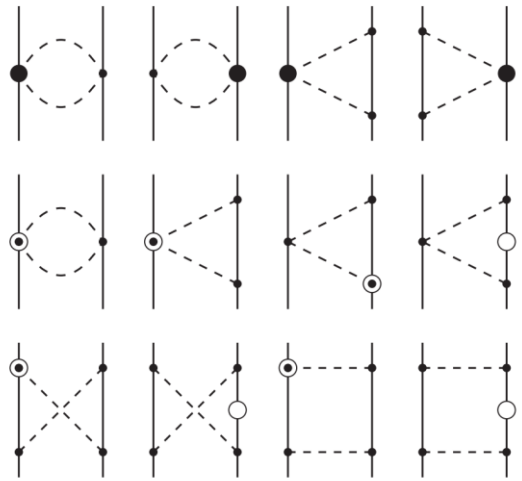


FIG.5. The 2π-exchange diagrams of order three.

This contribution to the 2PE is the crucial one, because it provides an intermediate-range attraction of proper strength.

Next-to-next-to next-to-leading order ($\mathrm{N^3LO}$), $\boldsymbol{\nu = 4}$

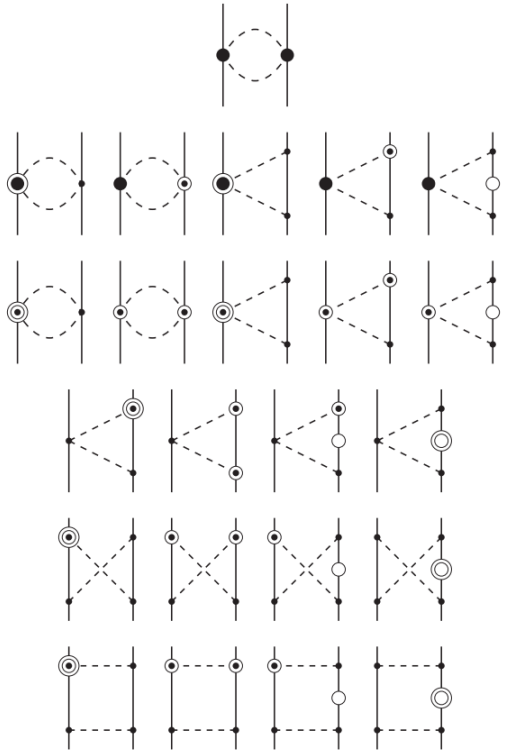


FIG.6. One-loop 2π-exchange contributions to the NN interaction at order four.

This order wishes to formulate this contribution such that relativistic elastic unitarity is satisfied.

Ref. [6] R. Machleidta, D. R. Entem, Chiral effective field theory and nuclear forces

Three-nucleon interactions (3NFs) from ChEFT

Nuclear three-body forces in ChPT were initially discussed by Weinberg

Next-to-leading order $(\mathrm{NLO}), \boldsymbol{\nu} = \mathbf{2}$

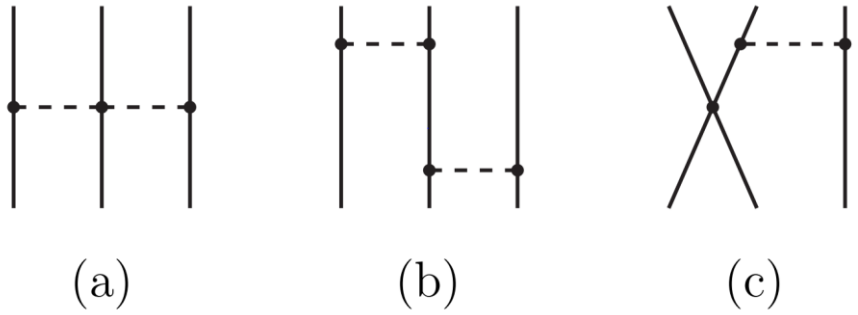


FIG. 7. Three-nucleon force diagrams at NLO.

- ν = 2, which is obtained for no loops (L = 0)
- The irreducible topologies of these graphs cancel against the recoil corrections from the reducible ones, leaving no net irreducible 3N
- There is no genuine 3NF contribution at NLO. The first non-vanishing 3NF appears at $\mathrm{N^2LO}$.

Next-to-next-to-leading order $(\mathrm{N^2LO}), \boldsymbol{\nu} = \mathbf{3}$

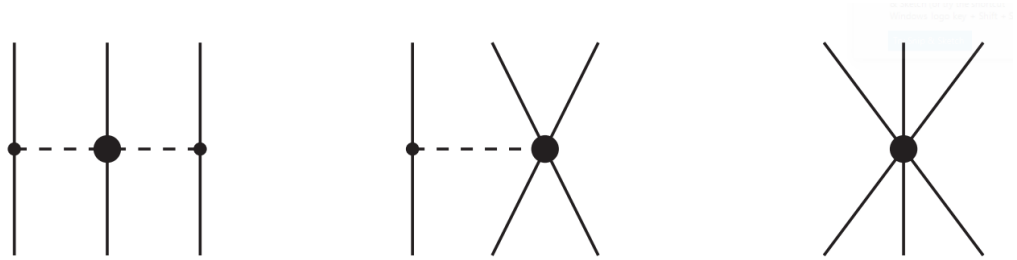
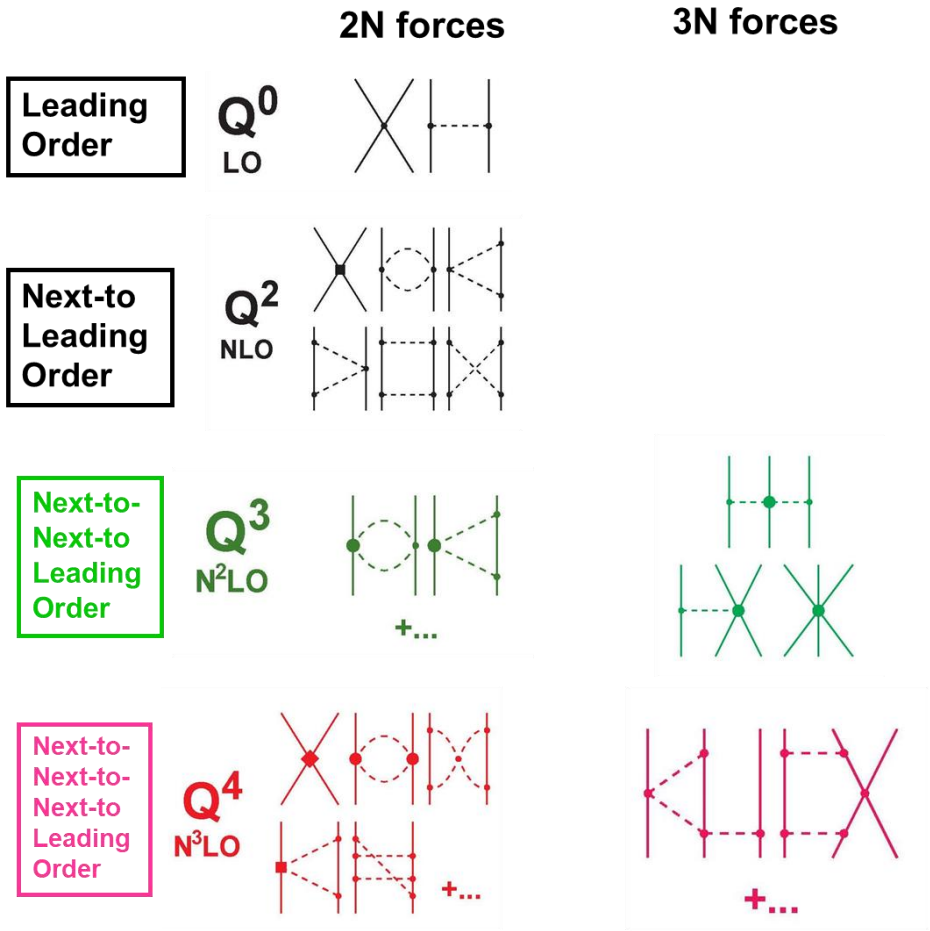


FIG. 8. The three-nucleon force at NNLO. From left to right: 2PE, 1PE

- $\boldsymbol{\nu}$ = 3, which is obtained when there are no loops (L = 0)
- There are three topologies which fulfill this condition, known as the two-pion exchange (2PE), 1PE

Ref. [6] R. Machleidta, D. R. Entem, Chiral effective field theory and nuclear forces

Summary of 2NFs and 3NFs from ChEFT



Ref. [6] R. Machleidta, D. R. Entem, Chiral effective field theory and nuclear forces

Back into this work

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Purpose

- The study of pairing in nuclei including three-nucleon forces.
- The development of low-momentum interactions based on renormalization group (RG) methods.

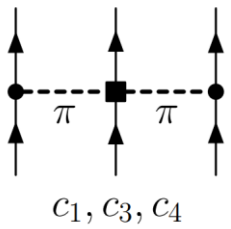
The RG plays a central role in EFT.

It determines the running of low-energy couplings.

I) Describing pairing

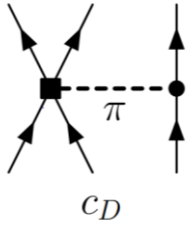
: Via a perturbative expansion around the HFB state translates into solving a generalized gap equation.

Consider the anomalous self-energy : a systematic study of pairing gaps generated using a pairing interaction at first order in chiral low-momentum interactions



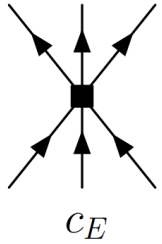
long-range
2π-exchange part v_{ci}

FIG. 9.



Intermediate-range
1π-exchange part v_D

FIG. 10.



Short-range
contact interaction v_E

FIG. 11.

Following this step:

1. calculations start from the N^{3}LO NN potential (EM 500 MeV) include the leading N^{2}LO 3N forces based on chiral EFT
2. construction an antisymmetrized, density-dependent two-body interaction $\bar{V}_{3N}$ to general isospin asymmetries:

$$\overline{V}_{3\mathrm{N}}(k_{\mathrm{F}}^{\mathrm{n}}, k_{\mathrm{F}}^{\mathrm{p}}) = \mathrm{Tr}_{\sigma_3,\tau_3} \int \frac{d\mathbf{k}_3}{(2\pi)^3}\, \theta(k_{\mathrm{F}}^{\tau_3} - |\mathbf{k}_3|)\, \mathcal{A}_{123}\, V_{3\mathrm{N}}\,.$$

For neutron matter, only the c_1 and c_3 parts of 3N forces enter [7], but with protons present, all parts contribute at the density-dependent two-body level.

Ref. [7] K. Hebeler and A. Schwenk, Phys. Rev. C 82, 014314 (2010).

I) Describing pairing

: Via a perturbative expansion around the HFB state translates into solving a generalized gap equation.

- The density-dependent two-body interaction $\bar{V}_{3N}$ corresponds to the normal-ordered two-body part of 3N forces [7]. Normal ordering with respect to a superfluid HFB state leads to a two-body pairing interaction

- where $\bar{V}_{3N}$ is added with a combinatorial factor 1 to the antisymmetrized NN interaction $\mathrm{V_{NN}}$

In this work, Focus on the $^1S_0$ partial-wave contribution that dominates isovector pairing , so that the first-order pairing kernel reads

$$\overline{V}^{^1\mathrm{S}_0}_{\mathrm{pairing}} = \overline{V}^{^1\mathrm{S}_0}_{\mathrm{NN}} + \overline{V}^{^1\mathrm{S}_0}_{\mathrm{3N}} \ .$$

I) Describing pairing

: Via a perturbative expansion around the HFB state translates into solving a generalized gap equation.

According to the first-order pairing kernel reads

$$\overline{V}^{^1\mathrm{S}_0}_{\mathrm{pairing}} = \overline{V}^{^1\mathrm{S}_0}_{\mathrm{NN}} + \overline{V}^{^1\mathrm{S}_0}_{\mathrm{3N}} \, .$$

It is convenient to separate the contributions to $\overline{\mathrm{V}}_{\mathrm{3N}}$ from the summation over occupied neutron and proton states.

$$\overline{V}_{\mathrm{3N}}(k_{\mathrm{F}}^{\mathrm{n}}, k_{\mathrm{F}}^{\mathrm{p}}) = \overline{V}_{\mathrm{3N},\langle \mathrm{n} \rangle}(k_{\mathrm{F}}^{\mathrm{n}}) + \overline{V}_{\mathrm{3N},\langle \mathrm{p} \rangle}(k_{\mathrm{F}}^{\mathrm{p}})$$

The Graph shows the 3N contributions are both repulsive, so that pairing will be weaker in nuclei compared to the NN-only level. In addition, $\overline{V}^{^1\mathrm{S}_0}_{\mathrm{3N},\langle \mathrm{p} \rangle}$ is stronger than $\overline{V}^{^1\mathrm{S}_0}_{\mathrm{3N},\langle \mathrm{n} \rangle}$.

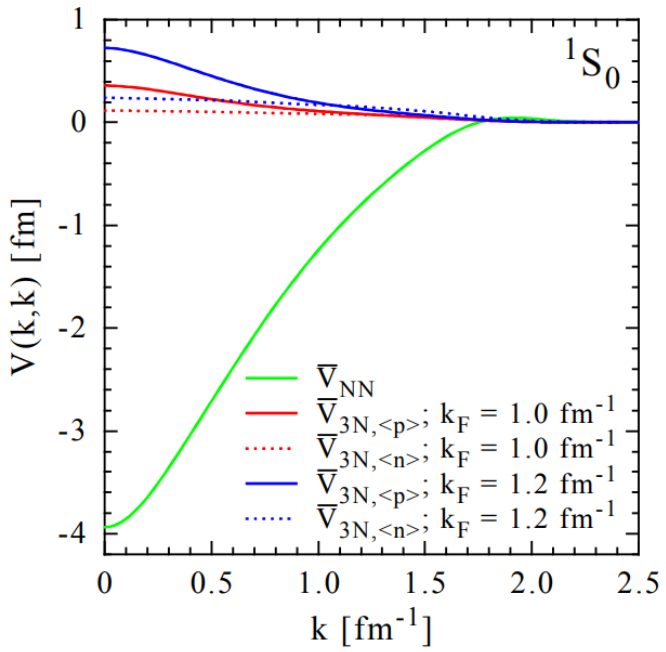


FIG. 12. Antisymmetrized momentum-space matrix elements for low-momentum NN and 3N interactions

II) Systematic EDF calculations of semi-magic nuclei

: Due to normal self-energy and drives the correlated single particle motion

- It is taken as a semi-empirical Skyrme parameterization.

It is convenient to develop an operator representation of $\overline{V}_{3\mathrm{N},\langle\tau\rangle}(k,k';k_{\mathrm{F}}^{\tau})$ We take a rank-m separable Ansatz of the form

$$\overline{V}^{^1\mathrm{S}_0}_{3\mathrm{N},\langle\tau\rangle}(k,k';k_{\mathrm{F}}^{\tau}) = \sum_{\alpha,\beta=1}^{m} g_{\alpha}^{\tau}(k)\,\lambda_{\alpha\beta}^{\tau}(k_{\mathrm{F}}^{\tau})\,g_{\beta}^{\tau}(k')$$

- local-density dependence based on the Campi-Bouyssy prescription for k_{F}^{τ} [8] leads to pairing gaps within 10 keV of the gap.

with m ≤ 4. The density dependence is parameterized as a polynomial in k_{F}^{τ}: $\lambda_{\alpha\beta}^{\tau}(k_{\mathrm{F}}^{\tau}) = \sum_{i=3,4}\lambda_{\alpha\beta,i}^{\tau}(k_{\mathrm{F}}^{\tau})^{i}$ Given our Ansatz, the functions $g_{\alpha}^{\tau}(k)$ and the coefficients $\lambda_{\alpha\beta,i}^{\tau}$ are fitted to the momentum-space matrix $\overline{V}^{^1\mathrm{S}_0}_{3\mathrm{N},\langle\tau\rangle}(k,k';k_{\mathrm{F}}^{\tau})$ for various values of k_{F}^{τ} from 0.6 to 1.6 fm^{-1}

Ref. [8] X. Campi and A. Bouyssy, Phys. Lett. B 73, 263 (1978).

According to Skyrme parameterization have been constrained from Hartree-Focksolve , After that, solving the HFB equation. (HFB state translates into solving a generalized gap equation.)

Next step:

III) Systematic calculations of the odd-even mass staggering generated

(Using a microscopic pairing interaction at first order in chiral low-momentum interactions.)

- These provides a measure of the pairing gap associated with the lack of binding of an odd isotope/isotone relative to its even neighbors.
- The three-point mass differences are computed in the exact same way from experimental data and EDF calculations.
- The calculations of odd-even nuclei are computed through the self-consistent blocking procedure performed within the filling approximation.

Result

The figure shows theoretical and experimental neutron/proton three-point mass differences $\Delta^{(3)}_{n/p}$ along isotopic/isotonic chains

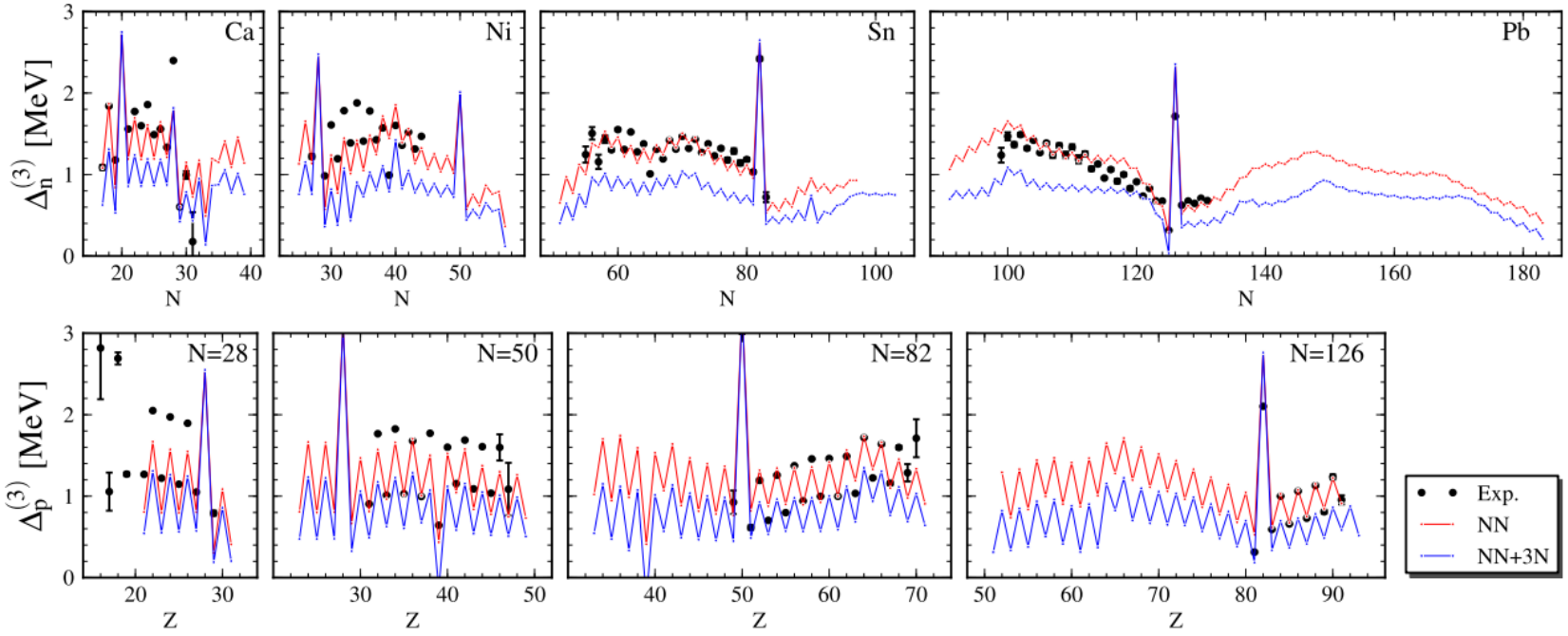


Fig. 13. Theoretical and experimental neutron/proton three-point mass differences $\Delta^{(3)}_{n/p}$ along isotopic/isotonic chains from one-proton to one-neutron drip-lines based on low-momentum NN and 3N interactions with Λ/Λ_{3NF} = 1.8/2.0 fm^{-1}.

(Note that for neutron-rich tin isotopes, the chemical potential is just below threshold such that the one-neutron drip-line is located well before the two-neutron drip-line.)

Analysis

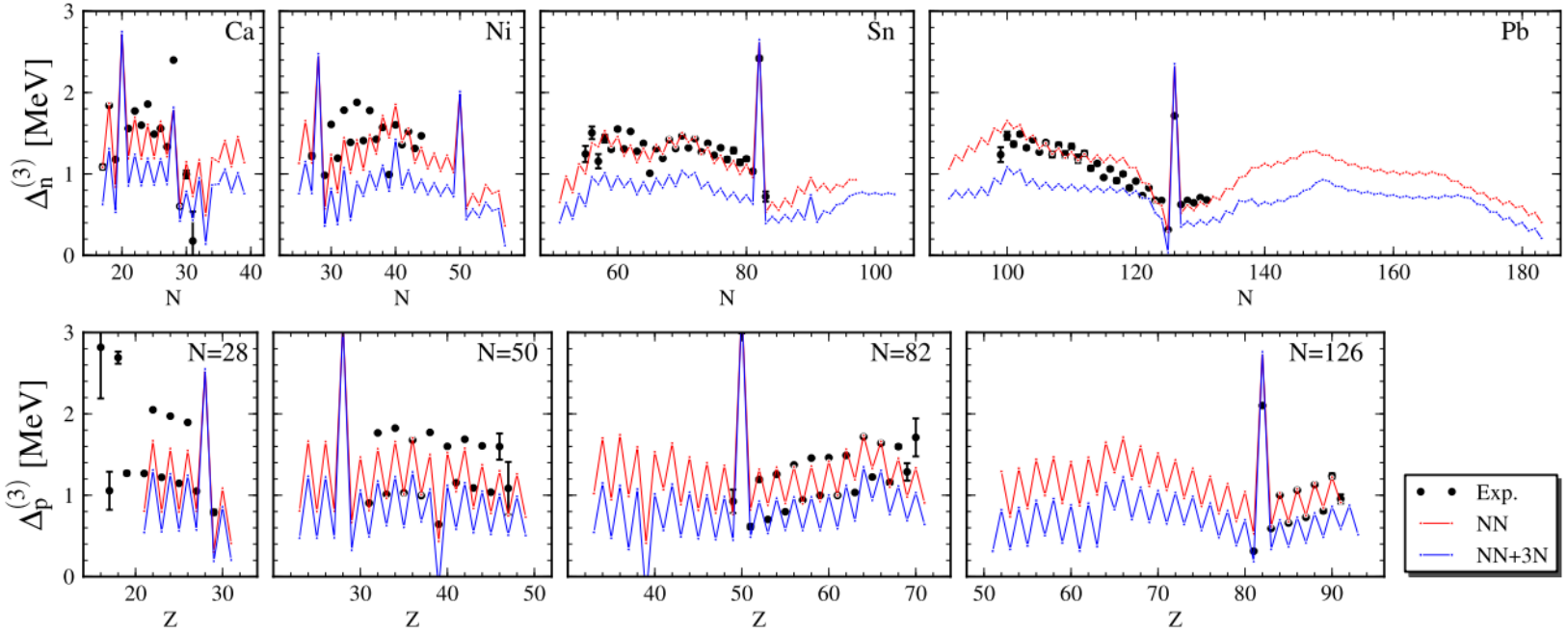


FIG. 13.

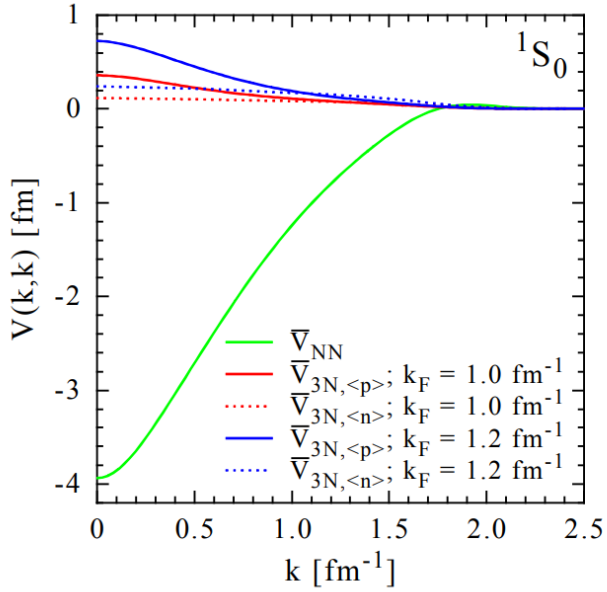


FIG. 12.

- Fig. 13. provides main results for NN only is that theoretical neutron and proton pairing gaps computed at lowest order are close to experimental ones for a large set of semi-magic spherical nuclei.
- The addition of the first-order 3N contribution then lowers pairing gaps systematically by about 30%.

IV) The studying of cutoff dependence of the pairing gaps.

(Given estimation of the theoretical uncertainties due to short-range higher-order NN and many-body interactions)

- Consider $\Delta^{(3)} - \overline{\Delta}^{(3)}$, where $\overline{\Delta}^{(3)}$ accounts for that oscillation and is obtained by treating the odd nuclei as if they had the same structure as the even ones [9].

➡ The pairing gaps at the NN-only level vary by ≈ 100–200 keV, and even smaller for less smooth cutoffs.

➡ When 3N forces are included, the theoretical uncertainties are of similar size, ≈ 100 – 250 keV, as indicated by the range of dotted and solid lines in Fig. 3

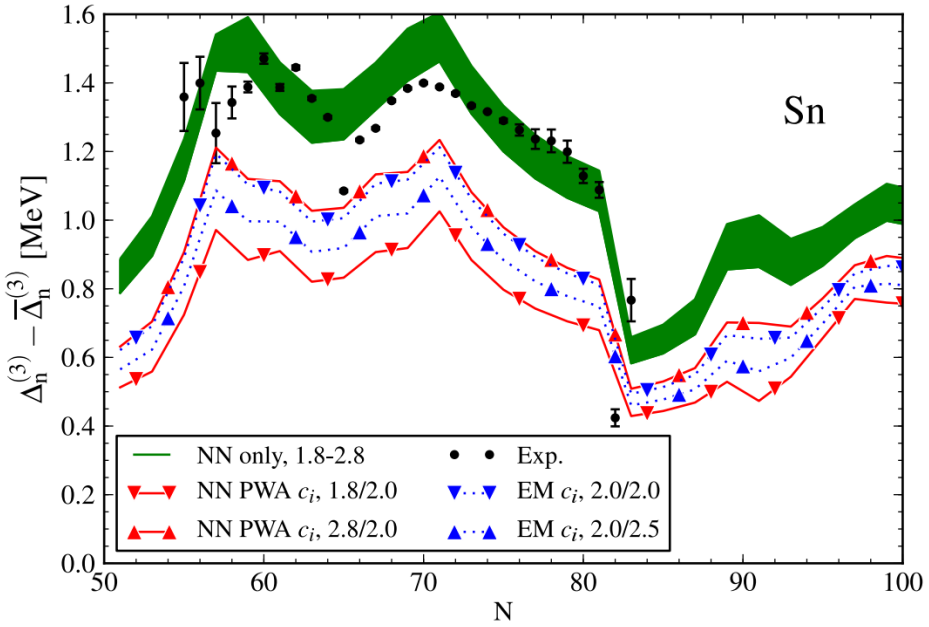


FIG. 14. Cutoff dependence of the three-point mass difference $\Delta_n^{(3)}$ along the tin isotopic chain, with the oscillating part $\overline{\Delta}^{(3)}$ subtracted.

Ref. [9] T. Duguet et al., Phys. Rev. C 65, 014311 (2002).

Conclusion

- It is necessary to include 3N contributions to the pairing interaction for a quantitative description of nuclear pairing gaps.
- With microscopic 2NF and 3NF effects, the nuclear magic number is reproduced.
- The up to NNLO, the results leave about 30% room for contributions from higher orders.
- The normal self-energy and higher-order contributions to the pairing kernel must be computed consistently based on low-momentum NN and 3N interactions.
- The empirical formula of EDF is described by the microscopic phenomenon from ChEFT.

Thank you

Renormalization group approaches

- The relevant details of high-energy physics for calculating low-energy observables can be captured in the scale-dependent coefficients of operators in a low-energy Hamiltonian.

⬇

It implies that we can include as much irrelevant coupling to incorrect high-energy physics as we want by using a large cutoff, with no consequence to low-energy predictions.

(assuming we can calculate accurately.)

Ref. : S. K. Bogner, R. J. Furnstahl and A. Schwenk, Prog. Part. Nucl. Phys. 65, 94 (2010).

How to do this EFT program?

This theory, one needs to assume some sort of perturbative expansion such that only a finite number of terms contribute at a given order. This expansion is provided by powers of small external momenta over the chiral symmetry breaking scale, $\Lambda_\chi \approx$ 1GeV

Take the following steps:

- Write down the most general Lagrangian including all terms consistent with the assumed symmetries, particularly, spontaneously broken chiral symmetry. (Note: There will be infinitely many terms.)
- Calculate Feynman diagrams. (Note: There will be infinitely many diagrams.)
- Find a scheme for assessing the importance of the various diagrams (because we cannot calculate infinitely many diagrams).

The Hierarchy of Nuclear Forces

Why we need Chiral perturbation theory (ChEFT) ?

- ChEFT and power counting imply that nuclear forces emerge as a hierarchy controlled by the power ν,

- In graphs are analyzed in terms of powers of small external momenta over the large scale: $(Q/\Lambda_\chi)^\nu$
- where Q is generic for a momentum (nucleon three-momentum or pion four-momentum) or a pion mass and $\Lambda_\chi \approx$ 1GeV is the chiral symmetry breaking scale

$$\text{Power} = -2 + 2A - 2C + 2L + \sum_{\text{all vertices}} \Delta_i$$

$$\text{with } \Delta_i = d_i + \frac{n_i}{2} - 2$$

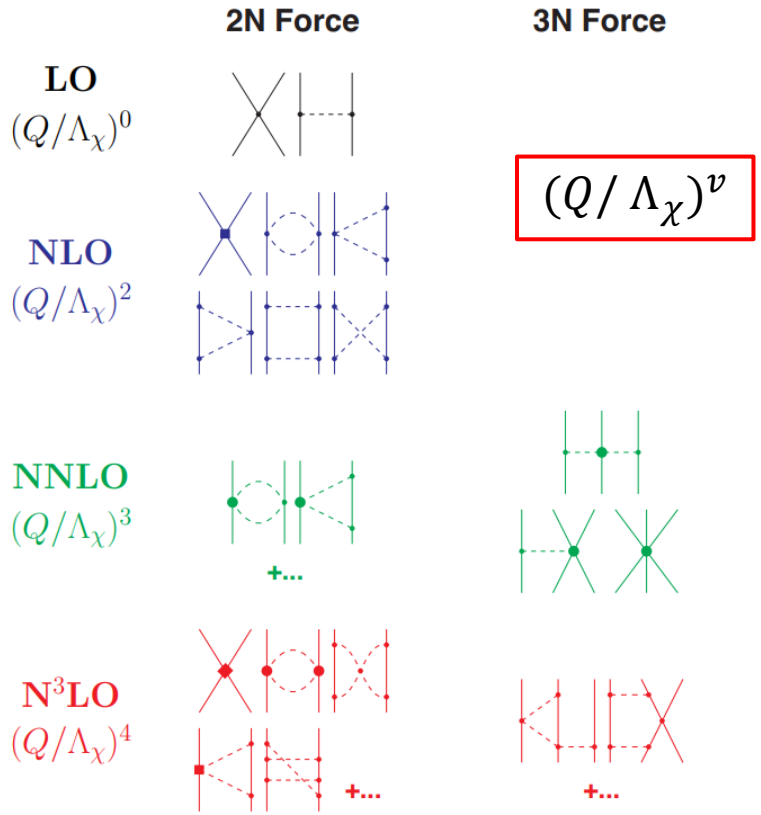


Fig.3. Hierarchy of nuclear forces in ChPT.

Ref: [5] R. Machleidt, Nuclear Forces - Lecture 5NF from EFT I (Pohang 2016)

The organizational scheme

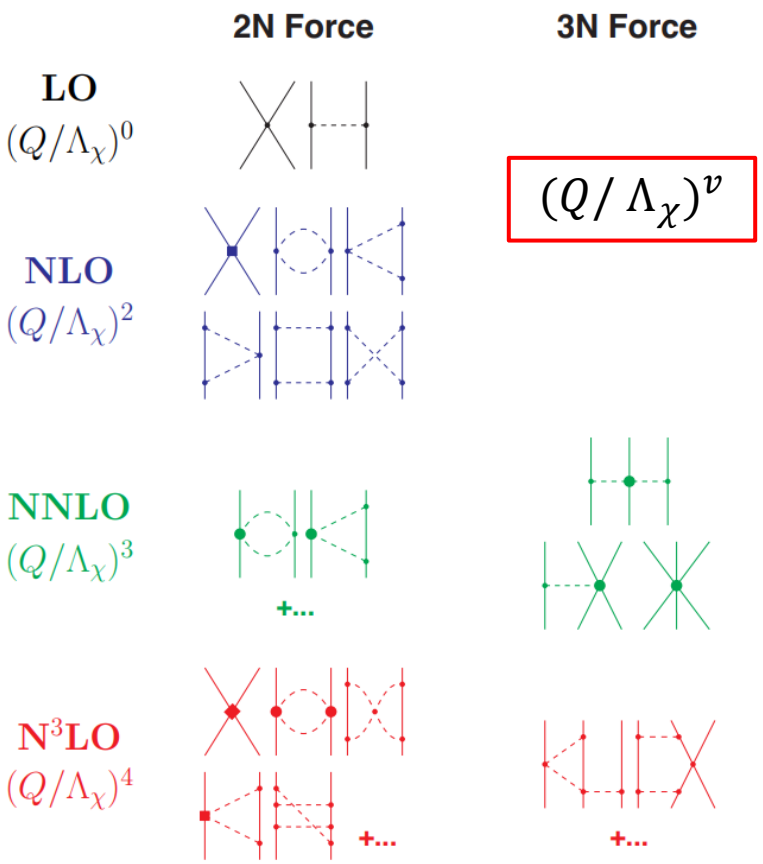
- Power Counting

$$(Q/\Lambda_\chi)^\nu$$

Organize the contributions in terms $(Q/\Lambda_\chi)^\nu$

Where Q denotes a momentum (derivative) or a pion mass (m_π);

$\Lambda_\chi \approx 1\ \mathrm{GeV}$, which is the chiral symmetry breaking scale, and $\nu \geq 0$.



$(Q/\Lambda_\chi)^\nu$

The Hierarchy of Nuclear Forces

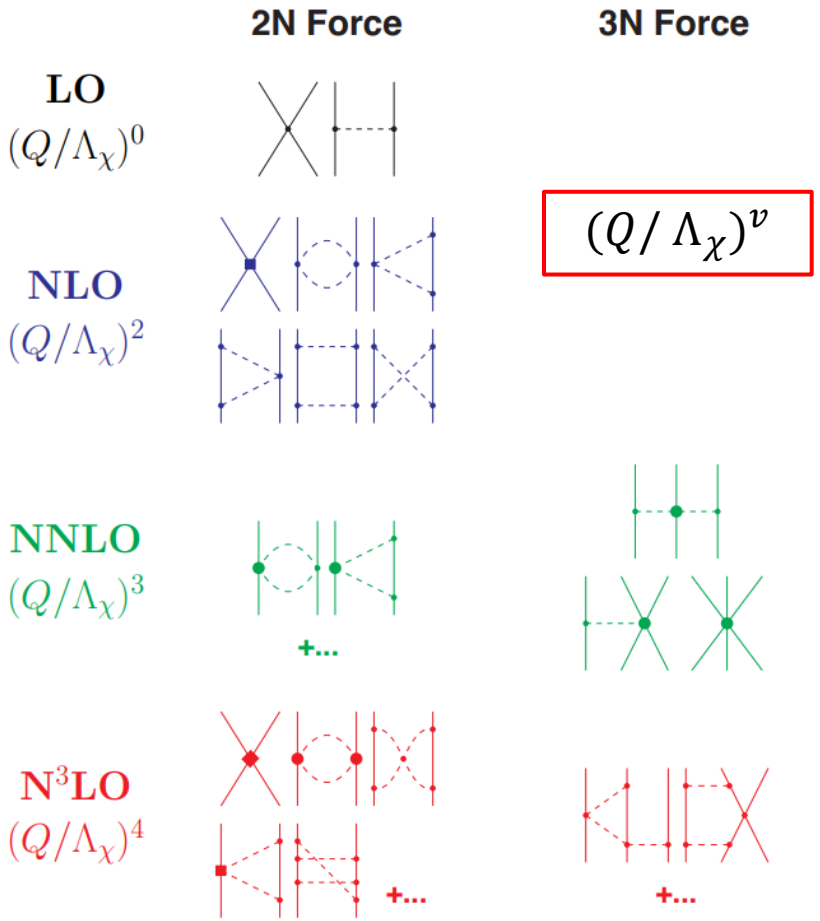
$$\text{Power} = -2 + 2A - 2C + 2L + \sum_{\text{all vertices}} \Delta_i$$

$$\text{with } \Delta_i = d_i + \frac{n_i}{2} - 2$$

Where $\boldsymbol{C}$ is the number of separately connected pieces.

$\boldsymbol{L}$ is the number of loops in the diagram; $\boldsymbol{d_i}$ is the number of derivatives or pion-mass (m_π) insertions.

$\boldsymbol{n_i}$ is the number of nucleon fields (nucleon legs) involved in vertex i; the sum runs over all vertices $\boldsymbol{i}$ contained in the diagram under consideration.



Ref: [5] R. Machleidt, Nuclear Forces - Lecture 5NF from EFT I (Pohang 2016)

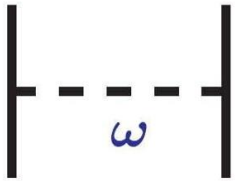
NN contact terms :

Why contact terms ?

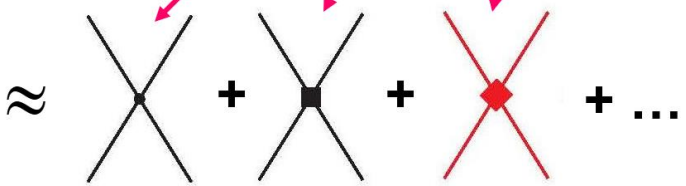
1. Renormalization: Absorb infinities from loop integrals.
2. Contact terms take care of the short range structures without resolving them.

- The successful test of the chiral pion-exchange contributions in peripheral partial waves
- It's an expansion in small momenta Q, too small to resolve structures like a $\rho(770)$ or $\omega(782)$ meson, because $Q \ll \Lambda_x \approx m_{\rho,\omega}$ But the latter relation allows us to expand the propagator of a heavy meson into a power series,

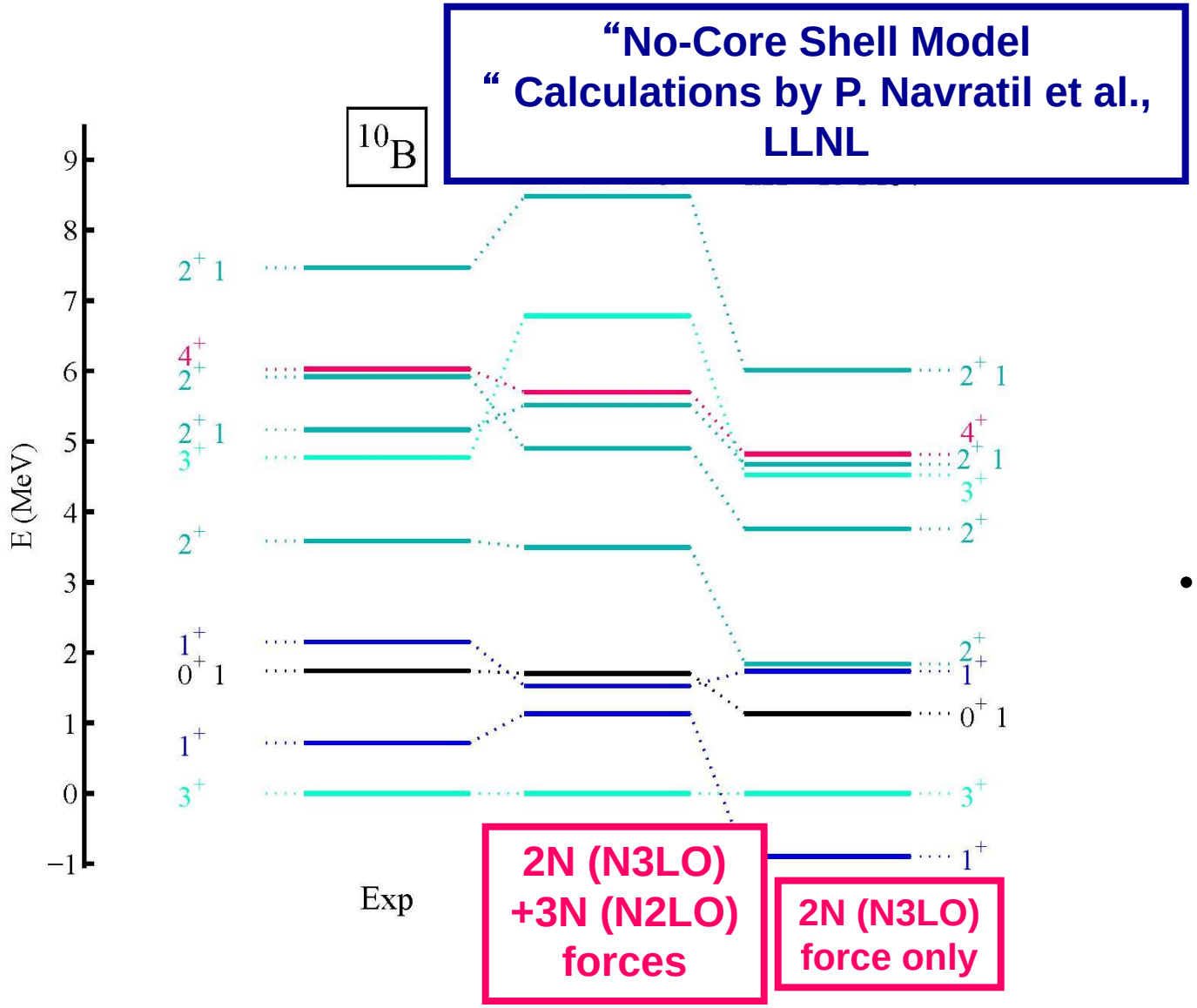
Consider the contribution from the exchange of a heavy meson



$$\frac{1}{m_\omega^2+Q^2} \approx \frac{1}{m_\omega^2}\left[1 - \frac{Q^2}{m_\omega^2} + \frac{Q^4}{m_\omega^2} - \dots\right.$$



Calculations applying



- Calculating the properties of light nuclei Using chiral 2N and 3N forces

- 3N forces play a key role in neutron-rich nuclei and matter in neutron stars, this also presents an important step to understanding pairing in neutron stars.